CS403

Lecture # 1:

Protect our failure => Robustness
Accurate work Provide => Efficiency
Updata data but not affect the privious data => Scalability

=>

Database

Organized collection of information in computerized format.

jis ka structure well-defined hu.

e.g; Cricket Players, Scholars, Movies, Food, Vehicale

=>

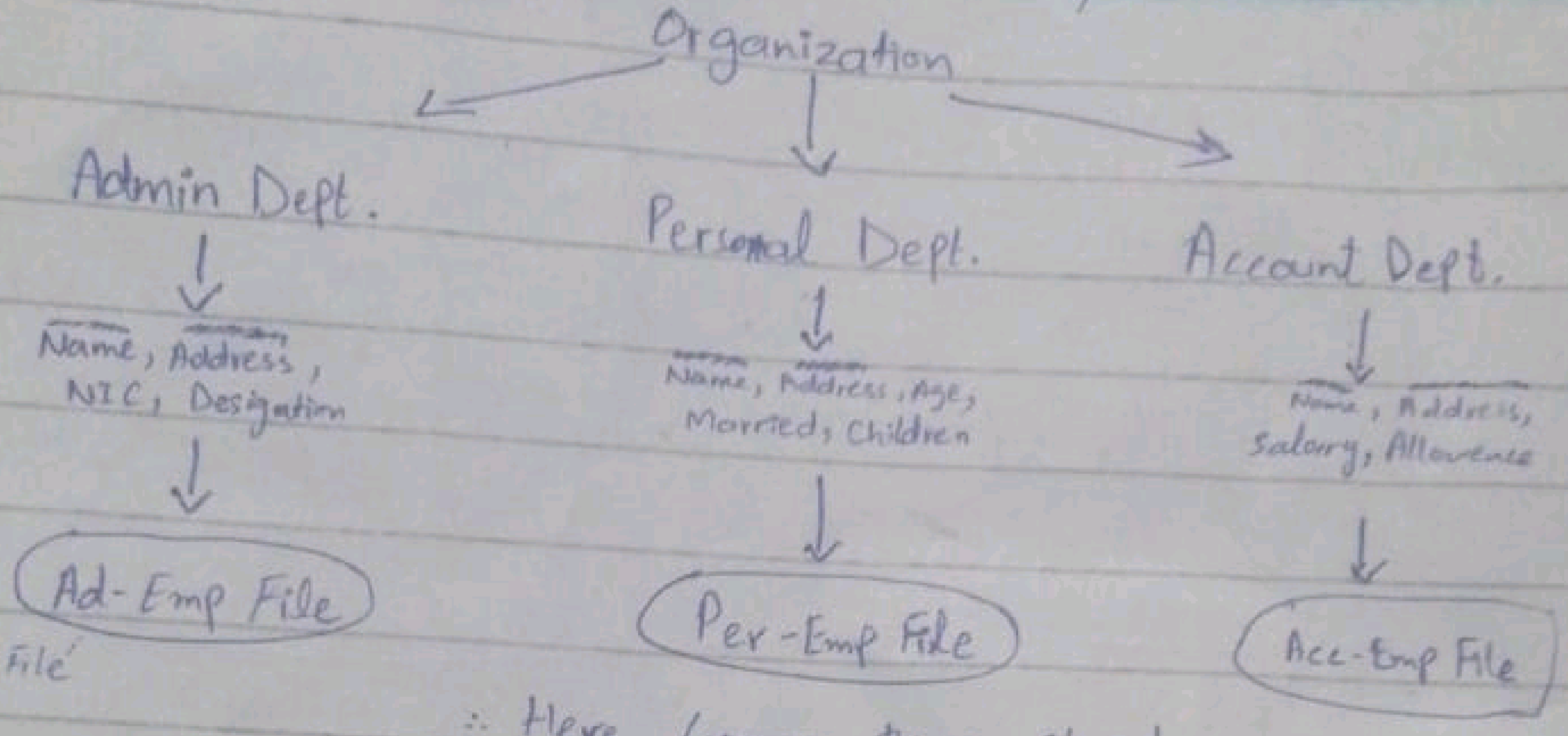
Importance of Database

In order to manage the commercial applications more efficiently databases are the ultimate choise society.

befor this file system.

=> Database & File Processing Systems.

Traditional File System


Organization
Admin Dept.
Name, Address, NIC, Designation
Ad-Emp File
File
Personal Dept.
Name, Address, Age, Married, Children
Per-Emp File
Account Dept.
Name, Address, Salary, Allowence
Acc-Emp File

∴ Here, common things stored Multiple times. So, It's False system.

As compared Database safe our time.

↓

Above Diagram except last line;

Database --- (Name, Address, NIC, Designation, Age, Married, Children, Salary, Allowence)

Difference b/w File & Database

Use separate data file for each application.	All application shares a pool of related & integrated data.
Data redundancy.	Minimal data redundancy.
Same data is recorded in several files.	Each occurrence of a data items is recorded only once.
Data inconsistency	Single version of data store.
Same update must be done in all occurance of same data item in each file.	Single update is required.
Difficulte to mainpulation data.	Easy to mainpulation data.

Advantages

- Data Sharing
- Data Independence
- Controlled Redundancy
- Better Data Integrity
  completeness, Accurate, efficient

Lect 2

=> Data (Raw facts & figures)

- Meaning less ✓
- Useless ✗

We use the data to made information.

=> Database Application:

is a Program or group of Programs which is used for Performing certain operation on the data

=> DBMS:

- Managemet of Data in the Database
- Managementof users associated with the database

=> Advantages of Database System

- Data Consistency: Consistent Data & consistent change.

اگر کسی time میں بہت سارے ڈیٹا کو اپنی need کے according کام

کر دینا

- Better Data Security: DBMS can Very efficiently check that which user is performing which action and accessing which part of data.

Private or Public as you want

- Faster Application Developement:
- Economy of Scale:
  Examination Record can be used by Attendance record
- Better Concurrency Control:
  Access of database from a number of points simultaneously. e.g, ATM machine
- Better Backup & Recording Facility:
  (Periodic backup)

★ Data as a Resource

Levels of Data:

- Real World Data; at which entities or objects exist in reality. e.g, Student

- Meta Data; schema for the real world data. kitni space ly ga data, khan store krna hu, size, location, which type value store. (data about data)

- Data Existence:
  Name kyo hy, kitni age hy
  Ali, 16 is data occurrance

Application Programmer:
Do cooding

=> End Users

- Naive User:
  Database user, having little knowledge of database.
- Sophisticated:
  Application use bhi krty hy and set bhi krty hain.

=> Data Base Administrator

- Data base bnany valy.
- Make only structure not do coding.

Let#3

Data Base Architecture

- Where data actually stored.

To describe the structure of database is called database Architecture.

Three-Level

1) External Level (view level)
   Frontend — programer ready krty ha
2) Conceptual level
   kitna table hugy, kitna size hu ga, — designer ready krty hy
3) Internal level (Physical level / schema
   here DB Administrator work (btaty ha ka table kahan ja kr store huga.)
4) Physical Data organization (Database)
   Data stored in the form of file.

Lecture #5

Momina

CS201
CS403
CS304
CS301
CS502
CS619

Database development Process

① Preliminary Study.
Rough sa idea ready krna — آ لے جیسا بنانا ہے

2) Requirment analysis:
Features ko bnany ka lya requirment collect Ab sochna ka kya mana thi bnya th us me kuch change tu nhi krna.

③ Database design:
Paper پر Ap na map ready kya

④ Physical Design:
Kis Format pa, kis jagah/location pr huga

⑤ Implementation:
Start coding — developers

⑥ Maintenance:
If error occourd the cover that error

Tools

DFD => Data Flow Diagram

Adv => Simple, hide complexity
=> Show direction of information, Create link
=> Flow of information

CS 403

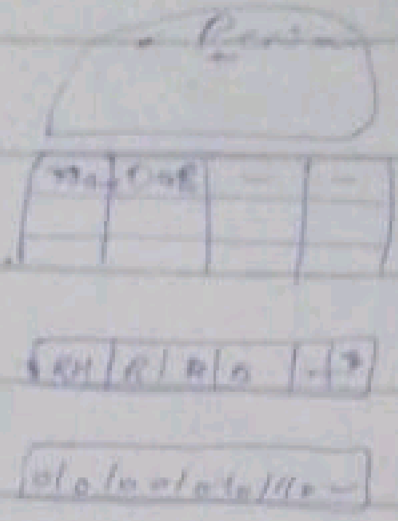
User

External views of Schema

Conceptual

Internal level

Physical view / Database



=> DBMS: Functions

- User Management
- Data management

=> Activities of DBMS

- Data Processing

Creation, Storing, Arrangement of data in database or data stores.

- A User accessible Catalog: Data Dictionary.

Stores all the Information of schema, user & rights of users and many more things about the database.

* Creation ادر کے Operations کی maintain hu rhi hoti hy is catalog mn.

- Transaction Support:

Used to Perform some manipulation on The data stored in the database. Like; CRUD

Create, Read, Update, Delete.

جب ہم Read / تبدیلی [illegible] Transaction ہو رہی ہوتی۔ [illegible]

آجائے تو DBMS حل کرتا ہے۔

- Concurrency Support: (DBMS Provide کرے to support)
  Two or more number of transactions to be executed simultaneously. All are ✓, no any information lost. (No lost info)

- Recovery Services:
  In case a database, to get corrupted due to any invalid action of someone, The DBMS should be able to recover itself to consistant state.

  Inconsistant state $\xrightarrow{\text{Processing}}$ Consistant state
  (minimum loss)

- Authorization Services: (DBMS Provide کرے)
  Number of users, who will perform a number of actions on The database. ([illegible])

- Data communication / Support for communication:
  DBMS should be able to communicate to the central database station.

- Integrity Services
  DBMS should allow the operation on The database which are real for the specific organization

DBMS Environments:

- Single User
- Multi-User

Different user intracting with database in different ways.

↳ Teleprocessing

DFD ⑤⑥

① Rules, ② Shape – name, work, ③ limitation

→ DFDs Flow of data b/w different processes.

→ DFDs are simple, & hide complexity.

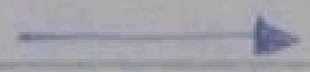
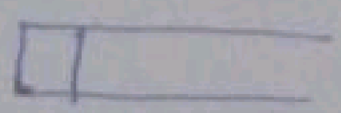
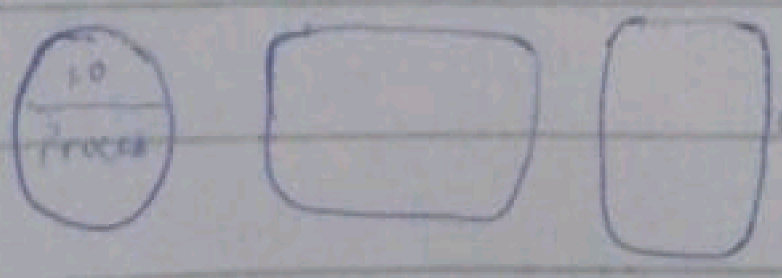
Descriptives [illegible] & Links b/w different processes describe the information flow.

○ Limitation

→ Don't provide us a way of desision Point.

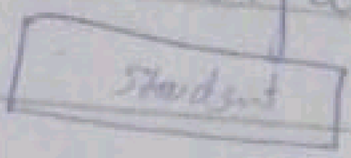
→ DFDs are focused on "flow of information only".

Symbds

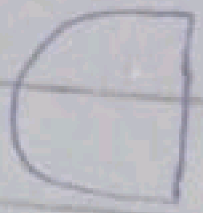
- Data Flow 
- Data Store 
- Process 

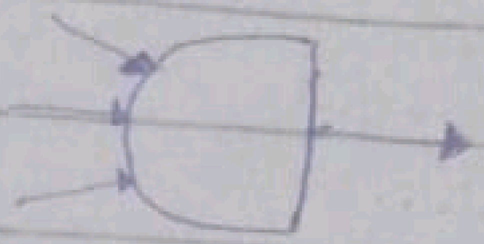
oval with round & rectangle

- External Entity => rectangular shape



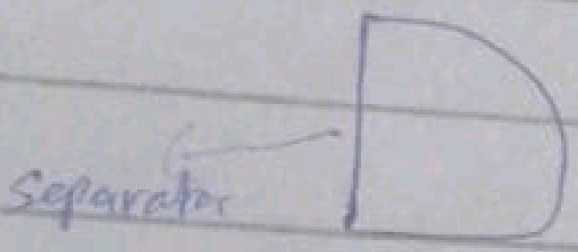
- Collector:

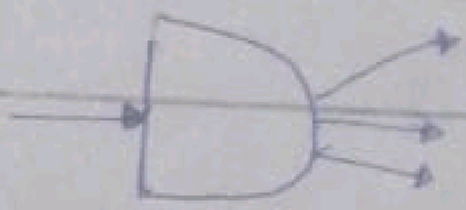




(Collector with multiple data-flow)

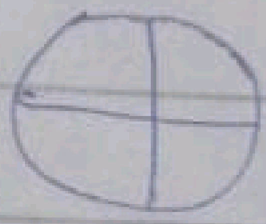
- Separator:

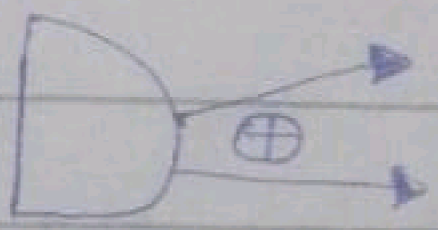




with multiple DF

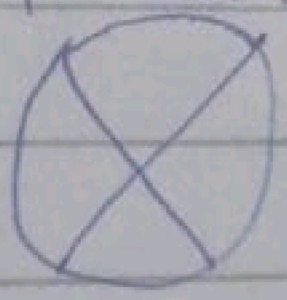
- Ring Sum Operator:

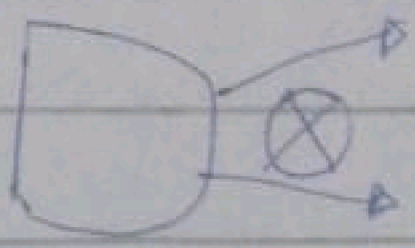




Separator with ring sum

- AND operator:





Types

- Content diagram
- level 0 diagram
- Detailed Diagram

① => Consist of single process and describe single system

② => Level 0 diagram in DFD is used to "describe the working of whole system". (Universty ka system)

=> Steps in Creating the DFD

1. Identify distinct modules of system
2. All modules one-by-one to show the internal functionality of the system
3. To make a Connection, Create links b/w entities.
4. Placing the number on Process

③ => Datailed DFD:

If we want more breifly explain then used this.

Data Dictionary

Here, Store the meta data.

Where store data — location, when, What, Which type all details in it.

Types of Data dictionary

- Integrated: used DFD

  In which stored, three level architecture schemas.

- Free standing:

  Tools which you used to make database design that information stored in ST.

Cross Reference Matrix

- Mark step by step when you complete your task.

Database Design Phase

- DB Design / Model:

  Logical structure of DB.

It's stored in DB schema — turn stored in Ddictionary.

- DB Modeling:

  Process of creating Logical structure of DB.

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E-R Model

Entity relationship data Model

=> graphical representation of the conceptual database design.

=> Constructs in E-R data Model

- Entity —— [Student] Object, characteristics
- Attribute —— [NoRdlA] Action, behaviour, Attribute column
- Relationship —— [Link] Relation/link b/w 2 Entities

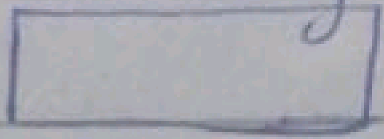
=> Different terms of Entity

- Entity type —— Entity ka name kya hy
- Entity Instance —— number of attributes
- Entity Set —— set of entities

student University

=> Classification of Entity type

- Strong Entity type —— e.g, Student

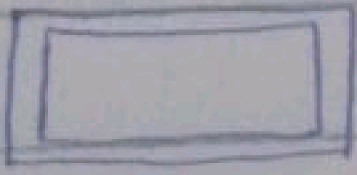


, No dependent & move by ownself.

- Weak Entity type —— e.g; Departement

Dependents on another entity & جو خود حرکت نہیں کر سکتی

Symbol :



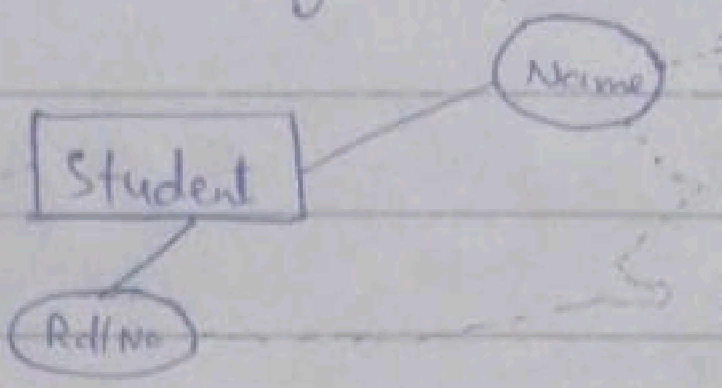
=> Attribute ovel

Characteristics/Column of Entity.

=> Domain of an Attribute:

Attribute type defined. domain [illegible]

Example:

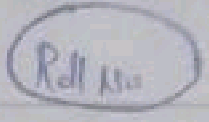


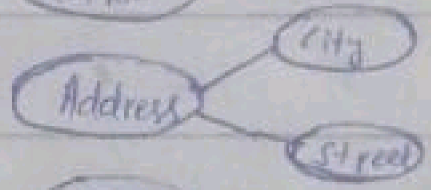
character

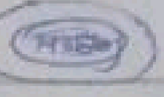
Entity

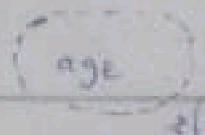
Attributes

=> Types of Attributes

Simple Attribute 

Composite Attribute 

Multiple Attribute  Hobbies

Derived Attribute 

جو کسی دوسرے Attribute اخذ کیا جائے

Lecture No8:

Key: that describe the attribute

1- Super key:

A set of two or more unique attribute ~~with~~ which taken collectively.

Multiple Attributes ko mil kr bn sakty ha.

2 - Candidate key:

All unique column/Attribute are called Candidate key. Jo prime key bnna ka lya hmesha ready hu. Sub super key.

ایک سے زیادہ ہو سکتی ہے

3- Primary: (No duplicate) (Unique) (Null x)
having unique value & no null value.
A candidate key chosen by the database designer to act as key is the Primary key.

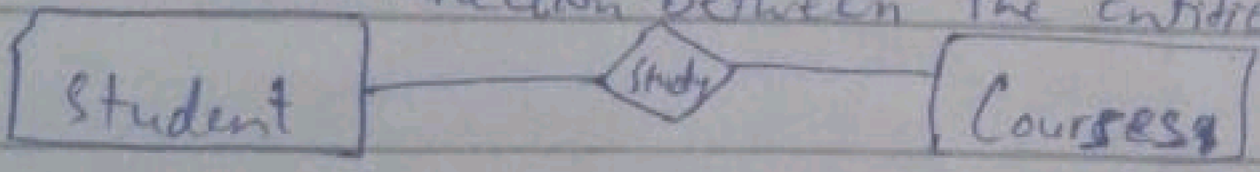
4- Alternate key:
Remaining candidates key → Alternate key

When it's become PK ?
when PK expired then we chosen by AK from CK.

5- Secondary key: (Duplicate) (Not unique)
Attribute jin ki value duplicate ho sakti hy.

Lecture No. 9

⇒ Relalationship:
Connection between the entities



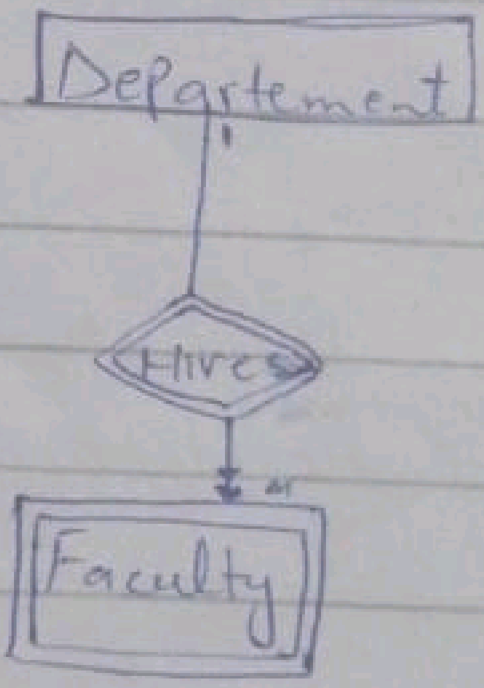
★ "Entities enrolled in a relationship are called its Participants".

=> Naming Relationships:

Itself relationship, when we have one entity.

=> Symbol for Relationships:

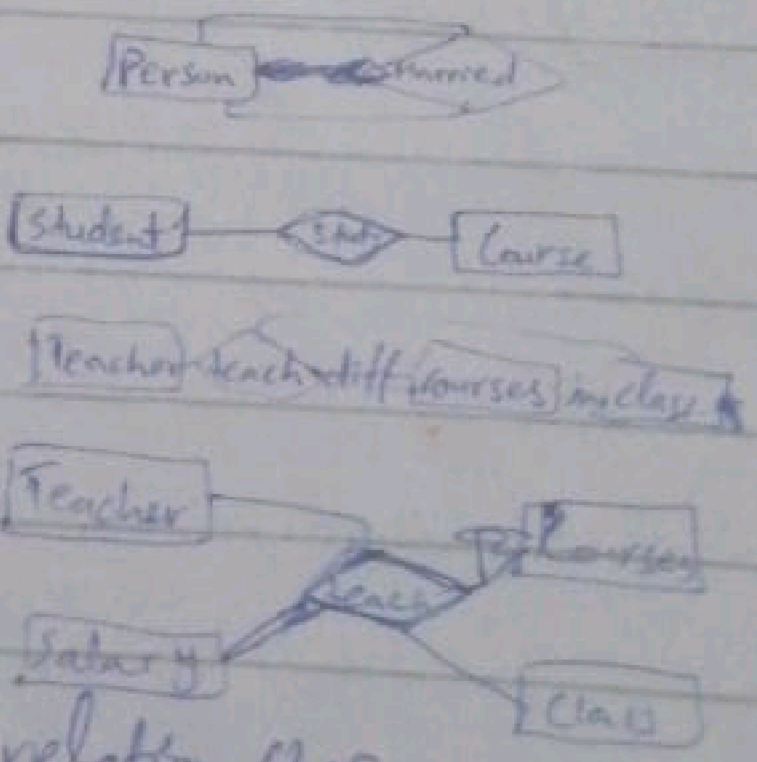
- Show as a Diamond.
- Diamond is doubled if one participant dependent on other
- Participants R connected by lines, Labeled — cardinality
- Total participation in indicated by double lines.


Departement
Hires
Faculty

Mara Allah
buht Mahrban
ha ♥

=> Types

1-	Unary	1 Entity
2-	Binary	2 Entities
3-	Ternary	3 Entities
4-	N-Array	N Entities


Person
married
Student
Study
Course
Teacher
teach diff
courses
in class
Teacher
teach
Courses
Salary
Class

No of entities = degree of relationship

=> Relationship Cardinality:

1 entity link with another entity how many times.

- One-to-One

e.g; Student have ID

- Many-to-One

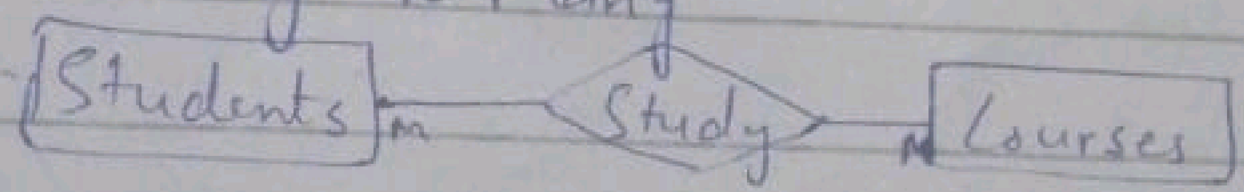
e.g; Student study in Department

- One-to-Many

e.g; Student study courses / Studen like Games

- Many to Many

e.g;



Lecture No 10:

Cardinality Represent

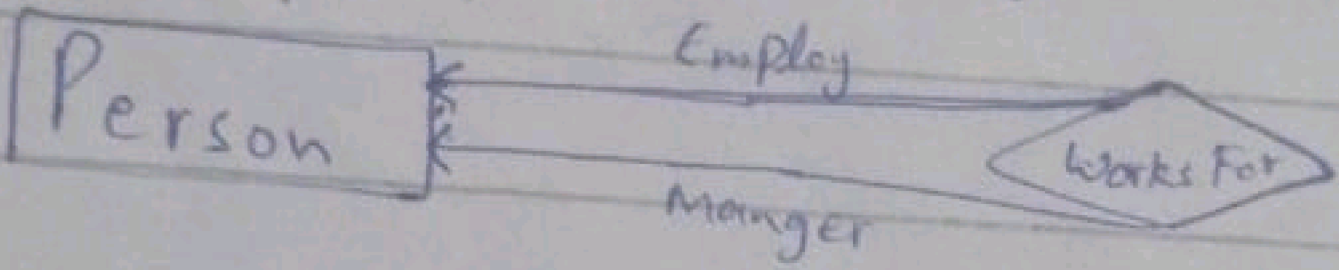
Shape	|	⋲
Arrow head notation	→	↠
Alphabetical notation	1.	M
Dot-based notation	1	●

=> Roles in Relationships

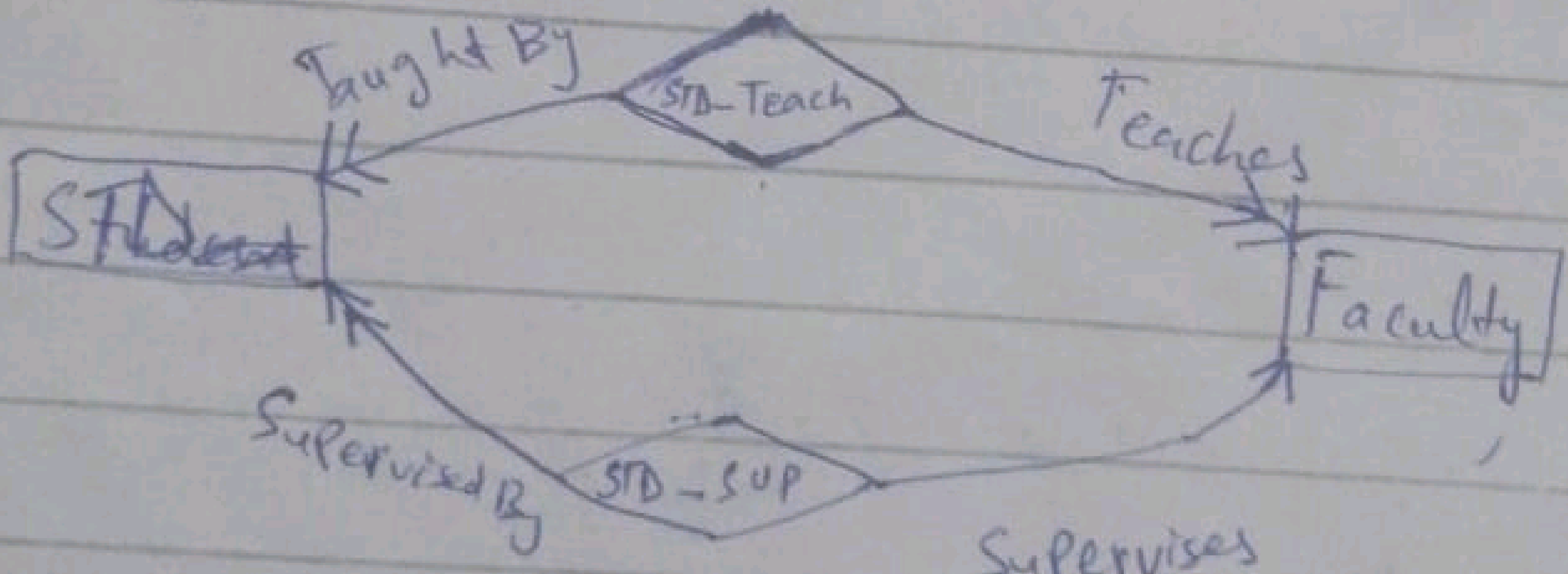
When an entity is involved in a relationship — This way — Role of entity in relationship.

=> Recursive Relationship.

- Roles in a unary relationship


Person
Employ
Works For
Manger

=> Multiple Relationships


Taught By
STD_Teach
Teaches
STudent
Faculty
Supervised By
STD_SUP
Supervises

Lecture No 11.

=> Inheritance

Transfer of the characteristics of class in OOP to other classes derived from it.

=> Supertype & Subtypes

Subtypes hold all the properties of their corresponding supertype.

=> Super & Subtype Relationship

- It allows us to create hierarchy of the entity
- we need not to write all the attributes again & again
- we can group similar type of entities & the ~~attrib~~ attribute at a certain/specific level.

=> Restriction with the help of Constraint :

(1) Specifying Constraint -

(2) Completeness Constraint

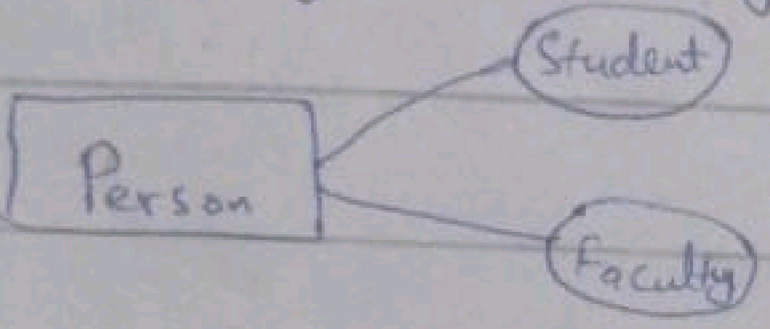
1- Total Completeness	2- Partial.
• All the attribute of supertype must be in subtype, then it's said to be total completeness	• May or may not all the attribute of super type must be in subtypes then it's Partial complete

③ Disjointness Constraints:

The rule defines the existence of a super type entity in a subtype entity.

- Disjoint constraint:

↓ restricts the existence of one instance of any supertype entity to exactly one instance of any of the subtype entity.



Person (Supertype ایک وقت میں کوئی ایک Property
کسی ایک میں ہی ہو سکتا ہے۔ ایک وقت دونوں میں
ایک جیسی خصوصیات نہیں پائی جاتی۔

Overlap Rule:

One instance of any supertype entity There can be multiple instance for more than one subtype entities.

Report ⟹ Column ki form ma huty hain

Lecture No 4:

=> Conceptual Database Design:

1- Developed in a semantic data model.
2- Free of data Model
3- Result from Analysis Phase
4- Graphical represented
5- More Expensive
6- Going to be transformed, and then implemented

=> Logical Database Design:

1- In Legacy data models
2- Free of Particular DBMS
3- Conceptual DB Design into anothe data model
4- Descriptive
5- Less Expensive
6- Going to be implemented

Data Model:

A set of constructs/tools used to develp a database design, generaly consist of 3 components:

Construct, manipulation Language & Integrity constraints

Relational Data Model

- Simplicity
- Strong Mathematical Foundation

Advantages

1- RDM has got a Precise meaning, So, there is no confusion.

2- If we want test something, we can it by Mathematical

3- (Relational algebra and relational caluclus) Provide 2 languages to solve math Problems

4- It Provide RDM certain boundaries. So, we can any modification or addition we want todo.

=> "An IBM Scientist E.F. Codd Proposed the $\underbrace{\text{relational data model}}_{\text{/Dominant data model/}}$ in $\boxed{1970.}$

=> Servers Jo Support krty hain RDM ko:

- IBM's DB2 family
- Informix
- Oracle
- SQL server
- Paradox
- Sybase
- Microsoft Access
- FoxBase

=> Why RDM?

For defining & designing external & Conceptual schemas,

- Based on single structure that is a relation.

★ RDM — 2-dimensional — table — row, column
Columns — attributes, field
(tuple, record
entity instance

★ Each cell of a table contains atomic single value.

★ Each column has unique name / entity type.

★ Value of the attribute / column come from the same domain

★ The order of columns is immaterial.

★ The order of rows is immaterial.

★ Each row / tuple / record is distinct, no 2 rows can be same

★ —

Mathematical relations

- Cartesian Product.

Apply more than 2 sets, in case of three sets, we will have a relation of ordered triplets.

Database relation

$A_1, A_2, A_3, \ldots A_n$ be some attributes and $D_1, D_2, D_3, \ldots D_n$ be their domains,

$$R = \{ A_1 : D_1, A_2 : D_2, A_3 : D_3, \ldots A_n : D_n \}$$

Lecture No 15:

=> Database & Math Relations

→ • The order of {columns} in mathematical relations does {matter},
In database relations it does {not matter}.

A × B ≠ B × A in Math

=> Degree of relation

Degree of relation = no. of columns/Attributes/field

=> Cardinality of relation:

" = no. of rows/tuples/records

=> Relation keys:

- Foregin key:

An attribute of a table B that is primary key in another table A is called foregin key.

=> Requirements,

- There can be more than zero, one or multiple foreign keys in a table, depending on how many tables particular table is related with,

- You can use the different names but remain in the same domain.

=> Represent

Primary key ———————— Solid line

Foreign key - - - - - - - - - - dotted line

=> Entity Integrity constraint:

Primary key is in table

=> Referential Integrity constraint:

Foreign key as a work ?

=> Significance of Constraints:

PK is a minimal identifier that is used to identify rows uniquely.

rows / records کو identify uniquely کے لیے use ہوتا ہے۔

=> Null Constraints:

A Null value of an attribute means that the value of attribute is not yet given, not defined yet.

آپ اس میں empty (rows) رکھ سکتے ہیں۔

Domain Constraint

=> ~~Default Value Constraints~~

This is essential constraint that is applied on every attribute, that is every attribute has got a domain.

=> Default Constraint:

If we don't give any value to any attribute, it will be give a certain (default) value.

=> Transforming Rules:

for converting conceptual database In logical database design.

Two Ways

- Manualy → Analyze & then transform
- CASE Tools — Automatically convert

=> Maping Entity types

Lecture #16:

Maping Relationships

- Relation is a structure
- Relationship is in between two relations of relational data model.

=> Binary Relationships

which are established between 2 entity type.

=> Three types of cardinality.

- One to many —— one instance link many instance
- One to One ←— P.k of one entity to other FK ~~pk~~
- Many to Many ←— Third table created (associative entity type)

=> Unary Relationship:

which involve a single entity. Recursive Relationship.

=> Super type & Sub type

- Common attribute are assigned to super type
- Specialized attribute are " " Sub types.

=> Data Manipulation Language (DML):

- Used to carry out different operations like Insertion, deletion, updation of data in database

=> Types

- Procedural Language:
  Tells the What to do, & How to do?
  e.g; Relational algebra.

- Non-Procedural Language:
  Only tells What to do? SQL is an example of

=> Relational Algebra:

- Operation work on one or more relations to define another relation.

- Allows expression to be nested, just as in arithmetic. This Property is called closure.

Unary

- 5 basic operations: Selection, Projection, Cartesian Product, Union & Set Difference.

Binary

- ? Data retrivel operation needed.

- It also has: Join, Intersection, Division operations.

Lecture No 17:

⇒ Uniary operation

- **Select operation:** (specific rows select)
  To select certain rows or tuples of a table, so it performs it action on the table horizentally
  - Greek Letter sigma σ is used to denotte.

$$\boxed{\sigma \overset{\text{rows}}{(\text{Condition})} (\text{Name of Table})}$$

- Project operation: (specific column select)

specific column کو ہم نے select کرنا rows کو نہیں۔
ہم نے کرنا choose

$$\Pi \left(\overset{c_1, c_3}{\text{column name which you want select}}\right) (\text{Table name})$$

⇒ Composition of Relational operation:

$$\pi_{c_1, \& c_3} \left(\sigma_{\{condition\}} (\text{Table name})\right)$$

⇒ Union:

All data select of both (2) Tables

- First table name ∪ Second table name

⇒ Intersection: (Common data selected)

- A ∩ B common data selec

⇒ Set difference:

Ka table (دوسرے) کا نہیں show data ہوتا جو Table کے:
display common data نہیں کرتا ہے۔

- First table name — Second table name

$\boxed{A - B}$

⇒ Cartesian Product:

1st table ki rows × 2nd table ki rows.

$\boxed{A \times B}$

⇒ Join Operation: (Binary Operation)

2 or more than 2 tables data selected from at the same time. Remove duplicate clm

Special form of cross product of two tables.

⇒ Types of Join

1- Theta join
2- Equi
3- Semi
4- Natural
5- Outer

T1 Left outer join T2
T1 right outer join T2

=> Theta Join

Cross Product : all the rows of first relation with all the rows of second relation.
only selected rows of a relation

$$\boxed{R \times \theta S}$$

Query:

first

$(\sigma_{\text{condition}}$ (Table name)) × Table name second

select rows

=> Equi Join

- All rows of T_1 multiply with All rows of T_2 but columns multiple with other Table common. PK — FK
- Common Attribute multiple of T_1 and T_2.
  ↓
  repeat appears on the new Table.

=> Natural Join

- most common & general form of Join.
- Join on the base of common attribute
  but here appers only one time.

=> Left Outer:

All the rows of left relation part of the output. r

=> Righ Outer:

All the rows selected of right relation output. Left ki common only.